Erfan Pirjahandideh

Applied AI & ML engineer · Computer vision research

Darmstadt, Germany | erfan@erfanpj.com
https://erfanpj.com | https://github.com/erfanpj | https://www.linkedin.com/in/erfan-pirjahandideh/

PROFILE

I'm a Computer Vision & AI Research Assistant at Fraunhofer IGD and an M.Sc. student in Artificial Intelligence and Machine Learning at TU Darmstadt (expected graduation [[MSC_GRAD_DATE]]). Before that I spent about eight years in telecom IT, leading a service desk and automating support work: 25% faster ticket resolution and 40% faster response times. I build things people can check: held-out evaluations, simple baselines and human approval before automated actions.

Open to applied AI, AI automation, AI enablement and ML engineering roles, and a master's thesis (brain and neurotech preferred). Berlin, Frankfurt or remote EU; based in Germany.

RESEARCH EXPERIENCE

Computer Vision & AI Research Assistant · Fraunhofer IGD, Darmstadt, Germany — Feb 2026 – present

- Work on attribute-driven identity synthesis; paper in preparation, target venue WACV 2027 (ongoing research, no published results yet).
- Build reproducible image-synthesis and evaluation pipelines with Python, PyTorch, CUDA and Stable Diffusion. Align faces, extract 512-dimensional embeddings and compare biometric error rates (EER, FMR/FNMR) across controlled experiments.
- Run experiments on A100 GPUs with Slurm; apply quality scoring and identity-wise selection, verify dataset integrity and document evaluation procedures.

SELECTED PROJECTS

Brain Signals Lab — EEG motor imagery · *Completed evaluation · public data* · github.com/erfanpj/brain-signals-lab

- Tested whether imagined left- vs right-fist movement can be decoded from the EEG of people the model has never seen: public PhysioNet data, 50 participants, 2,245 trials, five participant-held-out folds; band and settings chosen on training participants only.
- CSP + LDA reached 61.8% mean accuracy vs a 50.4% majority-class baseline (participant median 57.8%); a reduced EEGNet reached 53.9%. A static viewer shows every participant's result. Engineering result on one public dataset, not a clinical or neuroscience claim.

OpsDesk AI — human-approved support automation · *Local prototype · browser demo is a simulation* · github.com/erfanpj/opsdesk-ai

- Built a local support desk: retrieval over a handbook (keyword, BM25, dense, hybrid), structured proposals from a small local LLM, explicit human approval and mock execution.
- Approval is invalidated when a proposal changes; execution is idempotent per approval id; every step is written to a hash-chained audit log. A FastAPI review UI and an MCP adapter share the same service.
- On 60 held-out fictional tickets the rules baseline produced 37/60 fully correct proposals vs 25/60 for the local model, so I kept rules as the default.

Reachboard — Django workflow showcase · *Fictional data · browser demo is a simulation* · github.com/erfanpj/reachboard-showcase

- Django campaign workflow with server-authoritative pricing, ownership checks, CSRF-protected checkout, explicit state transitions, audit history and idempotent submission (a repeated submit adds no duplicate history).

Breast cancer classifier comparison · *Completed evaluation · public dataset* · github.com/erfanpj/Breast_Cancer_Project

- Rebuilt an earlier analysis with leakage-safe pipelines, nested cross-validation and a held-out test set scored once; automated tests check that no fit sees a test row.
- Tuned MLP F1 0.950 vs 0.951 for plain logistic regression on 143 held-out samples, so the simpler model is just as good. Not a clinical tool.

Exploratory Alzheimer's data analysis in Julia · *Descriptive analysis · synthetic data* · github.com/erfanpj/exploratory_alzheimers

- Descriptive analysis of a synthetic dataset (2,149 records) with validation against the data card, recorded cleaning counts, provenance notes, 95% intervals and 49 tests. Results describe the data generator; no clinical claims.

I used AI coding assistants on these independent projects; each README states what is fictional, local-only or simulated.

PROFESSIONAL EXPERIENCE

QA Team & Project Lead | Senior Developer Expert · NAK (MCI project, telecommunications), Tehran, Iran 2022 – 2024

- Led the implementation of a monitoring system covering more than 30 services for the mobile operator MCI (Hamrah-e Aval).
- Built a Chrome ticketing extension (jQuery, HTML, CSS) that reduced ticket-resolution time by 25%.

Service Desk & Service Management Team Leader · Behsa (MCI project, telecommunications), Tehran, Iran 2018 – 2022

- Set up and led a seven-person service desk handling more than 2,000 tickets a month; integrated MCI ticketing with Jira.
- Introduced ITIL-aligned workflows that improved response times by 40%; coordinated incident and request resolution against SLAs.

First Technical Support & Helpdesk Engineer · AHS (MTN project, telecommunications), Tehran, Iran 2016 – 2018

- Automated daily management reports for senior management and the CIO, improving report accuracy by 30%.
- Built Chrome extensions and service-desk dashboards that reduced manual workload by 20%; built Telegram bots for automated report updates.

Freelance Software Developer & Web Designer · Independent 2005 – present

- Build web applications, responsive websites, dashboards and workflow automations with PHP, WordPress, Joomla, JavaScript and Python, using AI-assisted development (Cursor, ChatGPT/Codex).

SKILLS

Applied AI & automation: LLM-assisted workflows, retrieval (keyword, BM25, dense, hybrid), local LLMs, MCP, human-in-the-loop approval, audit logging, FastAPI; AI-assisted development with Cursor and ChatGPT/Codex. Currently learning AI agents, LLM evaluation and workflow-automation platforms.

ML & computer vision: PyTorch, CUDA, OpenCV, scikit-learn, Hugging Face Diffusers and Datasets, Stable Diffusion, LoRA, InsightFace, ArcFace, ElasticFace, CosFace, CR-FIQA, ONNX Runtime.

Evaluation & research engineering: Held-out and nested cross-validation, simple baselines, EER and FMR/FNMR, bootstrap intervals, reproducible pipelines, identity-wise data selection, dataset-integrity checks, experiment documentation.

Software & infrastructure: Python, Julia, SQL, JavaScript, PHP, Groovy; Django, Celery, Laravel, PostgreSQL, MySQL, APIs; Git, Linux, Docker, Slurm, A100 GPUs, Playwright, Power BI.

Process & delivery: ITIL, Jira, SCSM, SLA/OLA, QA and software testing, workflow automation, stakeholder coordination, team leadership.

EDUCATION

M.Sc. Artificial Intelligence and Machine Learning · TU Darmstadt, Germany 2024 – present

In progress; expected graduation [[MSC_GRAD_DATE]].

B.Sc. and Associate Degree in Software Engineering · Islamic Azad University, Iran 2006 – 2013

Earlier AI master's coursework: 20 credits, no degree awarded.

ADDITIONAL

Publication: Co-author, "Personality Recognition from Text Based on Deep Learning with Recurrent Neural Network." 5th International Conference on Fundamental Research in Computer Engineering and Information Technology, 2017.

Award: MTN CEO Special Award for Innovation, April 2018.

Training: ISTQB software testing, DevOps Foundation (2021); ITIL Foundation, Python Foundation (2017).

Languages: English: advanced · Farsi: native · German: A1